

1. **Vector functions-Curves-I&II:** Vector functions and their domains, limits, continuity, differentiability, and integrals; parametric curves and the definitions of regular, simple, closed, and piecewise regular curves drawing parametric curves, tangent line to a regular curve, and the arclength of a curve; the arclength reparametrization, curvature, normal and binormal vectors, the osculating, normal, and rectifying planes, and the Frenet-Serret frame, the motion of objects in space.
2. **Domain-Limit-Continuity:** Functions of several variables: domain, level sets and graphs, limits and continuity.
3. **Differentiability-I&II:** Functions of several variables: partial derivatives, differentiability, the tangent plane to a graph, linear approximation near a point, the chain rule for the composition of two differentiable functions, and the directional derivative; higher order derivatives
4. **Parametric surfaces-I&II:** Definitions and properties, regular surfaces, cylindrical and spherical coordinates; tangent plane to a parametric surface, tangent vector to a level curve, tangent plane to a level surface.
5. **Maximum-Minimum-I&II:** Unconstrained optimization: points of local/global maximum/minimum, a necessary condition for local maximum/minimum, a sufficient condition of optimality; optimization under one equality constraint: the method of Lagrange multipliers in $\mathbb{R}^2$ and $\mathbb{R}^3$.
6. **Multiple-Integrals-I-IV:** The double integral in a rectangular domain; evaluation of double integrals over non-rectangular domains. change of variables in double integrals, evaluation of a double integral through the use of polar coordinates; evaluation of triple integrals over rectangular and non-rectangular regions; change of variables in triple integrals via cylindrical and spherical coordinates.
7. **Vector-Calculus-I-V:** Conservative vector fields, line integrals of scalar-valued functions; line integrals of vector fields; fundamental theorem of calculus for line integrals, path independence in connection with relationship with conservative fields; curve orientation, Greens theorem; curl and divergence of a vector field, orientation of a surface, surface integrals, the Stokes and divergence theorems.